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RESEARCH ARTICLE

Experimental Analysis of Trustworthy In-Vehicle Intrusion Detection System Using eXplainable Artificial Intelligence (XAI)

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ABSTRACT Anomaly-based In-Vehicle Intrusion Detection System (IV-IDS) is one of the protection mechanisms to detect cyber attacks on automotive vehicles. Using artificial intelligence (AI) for anomaly detection to thwart cyber attacks is promising but suffers from generating false alarms and making decisions that are hard to interpret. Consequently, this issue leads to uncertainty and distrust towards such IDS design unless it can explain its behavior, e.g., by using eXplainable AI (XAI). In this paper, we consider the XAI-powered design of such an IV-IDS using CAN bus data from a public dataset, named “Survival”. Novel features are engineered, and a Deep Neural Network (DNN) is trained over the dataset. A visualization-based explanation, “VisExp”, is created to explain the behavior of the AI-based IV-IDS, which is evaluated by experts in a survey, in relation to a rule-based explanation. Our results show that experts’ trust in the AI-based IV-IDS is significantly increased when they are provided with VisExp (more so than the rule-based explanation). These findings confirm the effect, and by extension the need, of explainability in automated systems, and VisExp, being a source of increased explainability, shows promise in helping involved parties gain trust in such systems.

INDEX TERMS Automotive, intrusion detection system, machine learning, deep learning, XAI, trustworthiness.

I. INTRODUCTION

An in-vehicle intrusion detection system (IV-IDS) is a software system employed by vehicle manufacturers to detect cyber attacks in automotive vehicles.

Each vehicle contains a set of in-vehicle networks (IVN), via which electronic control units (ECU) communicate

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internal vehicle functions. Attackers target these ECUs in an attempt to compromise important vehicle functions, such as showing the wrong speed on the dashboard, or disrupting acceleration when the accelerator pedal is pushed down [2] (such attacks are possible due to security vulnerabilities within the IVNs themselves [3], [4], [5], [6], [7]). The most widely used communication protocol in IVNs is the controller area network (CAN) [2], [8]. It is a high-speed, low-cost protocol with short data lengths (11 or 29 bits ID with 8 byte data

field), and messages (CAN frames) require no permission by a bus arbiter, which provides fast reaction times between connected devices [9]. Two critical vulnerabilities that can expose the CAN protocol (and by extension the ECUs) for attacks are, (i) CAN frames with low IDs have priority on the CAN bus (leading to denial of service (DoS) attacks), and (ii) there is no message authentication (leading to ID fabrication) [10].

One¹ way to design an IV-IDS is to utilize artificial intelligence² (AI), a method which falls under the class of IDSs known as anomaly-based, which, contrary to signature-based IDSs (matching data on the IVN against known attacks in a database), use anomaly detection algorithms to detect attacks on the IVN [16]. Anomaly-based IDSs are better at detecting zero-day attacks (attacks that have not yet been found by the vehicle manufacturer), however, they have higher false alarm rates compared to signature-based IDSs [17], and they have a tendency to be more complex, especially when they use AI. This highlights two problems: (i) An IDS that generates false alarms can lead to *distrust* and uncertainties when handling alarms [18], leading to the implementation of more overhead functions in attempts to increase the robustness of the system [19]. (ii) A complex AI-based IV-IDS has low *interpretability*, i.e., the AI model's mapping from input to output is hard to understand, or in other words, the reason why an alarm is generated (or not generated) is hard to understand. Model interpretability is an inherent challenge with complex AI models [20], [21], especially in ensemble learning (EL, e.g., RF) and deep learning (DL, e.g., DNN), which can contain millions of parameters in different mathematical structures. Moreover, interpretability is a trait that is argued to be of importance when ascertaining various other valuable AI model desiderata, such as *trustworthiness*, causality, and robustness [22], [23].

The previously mentioned challenges—handling false alarms and complex AI model structure—can be mitigated by finding ways to better *explain* AI models and their predictions. *Explainability* can be achieved by designing and using more interpretable AI models, or by creating tools that can explain the behavior of complex models post-hoc [24]. This issue is outlined in the growing [24] field of explainable AI (XAI, also called interpretable ML [25]) (a field that has gained even more traction since the general data protection regulation (GDPR) provided a right of explanation of systems based on automated processing [26], [27]). By providing explanations, an expert can gain a more holistic view of the AI model, generate new insights, and understand *why* the model works the way it does. Through this, better decisions can be

made in response to alarms, and the *trust* in complex AI models (and by extension AI-based IV-IDSs) can be increased.

Trust is the foundation that allows acceptance of new technologies [19], and explanations, in turn, are guarantees of trust when something is out of the ordinary, critical or surprising. Humans are very much at the center of XAI, where the concept of trust—something that occurs between humans—is extended to include trust between humans and automated systems [19]. However, the degree to which a human trusts an explanation is affected by cognitive ability, alertness, implied knowledge, available time, and of course the quality and truthfulness of the explanation itself. In the end, trust removes the need for overhead control functions that ensure system reliability—with enough trust in the system, it is simply not needed to the same extent [19]. This is what it means for an automated system to be trustworthy.

A. PREVIOUS WORK

Explanations in the field of XAI come in many flavors, and several researchers have begun working on taxonomies in order to organize the field [21], [22], [23], [24], [25].

In [28], Israelsen and Ahmed discussed “...trustworthy and explainable learning and AI, ethical and transparent autonomy, and safety-/user-aware intelligent systems...”, and emphasized the need for *assurances* in technology. Assurances are capabilities and methods that are designed into autonomous and intelligent systems, that help designers, users, and other stakeholders achieve *appropriate levels of trust* in said technology.

Lundberg et al. [29] propose SHAP, an influence- and visualization-based XAI method [24] that computes feature importance based on Shapley values from game theory. They evaluate the explainability of SHAP values by conducting a human study, where they compare human judgment with generated SHAP values. They found that there was a strong correlation between SHAP and human explanations.

The following related studies concern IDS research that utilizes AI methods and explainability, but the experiments are not conducted in the automotive domain with IVNs.

Mane et al. [30] propose a framework for explainable AI models for intrusion detection using the KDD dataset. They address the machine learning pipeline at three important stages (1) the data scientists, (2) the analysts, (3) the consumers. They use three algorithms from IBM's AI explainability 360 (AIX 360), and use different XAI models for different parts in the ML pipeline. They use SHAP, LIME, Contrastive Explanations Method (CEM), ProtoDash and Boolean Decision Rules via Column Generation (BRCG). They conclude that they provide satisfactory explanations at every stage of the ML pipeline, however, they do not produce any metrics evaluating the explainability (or trustworthiness) of the pipeline.

Wali et al. [31] propose an IDS with a random forest classifier using explainable AI (SHAP) to evaluate predictions. SHAP also gives insight into the model, as well as providing a stronger IDS. Predictions with low credibility are run through

¹Examples of IDS designs: measuring voltage levels of ECUs to detect anomalies [11], applying rolling windows across intervals of CAN traffic and using machine learning (ML) to classify CAN traffic [12], finding anomalies in the survival rates of malicious frames by analysing the ID fields [13].

²Global spending on cognitive and AI systems is forecast to reach \$77.6 billion in 2022, more than three times the \$24.0 billion forecast for [14]. Statista expects worldwide AI software market revenues to reach around \$126 billion by 2025 [15].

a second set of random forest classifiers. Their results show a 100% robustness with adversarial attacks using the CI-CIDS dataset. They emphasize that their proposed IDS provides transparent decision-making that enhances user trust, however, there is no metric that evaluates that claim.

Islam *et al.* [32] use domain knowledge to simplify the feature set of a dataset in order to make it more explainable. They compare different ML models, including random forest, which produces the best results. They use CIA (confidentiality, integrity, and availability) principles to augment/re-create the dataset (CI-CIDS2017). They conclude that their use of domain knowledge succeeds in raising the explainability, however, they do not provide metrics to support their claim.

Mahbooba *et al.* [33] trains a decision tree model to detect malicious data using the KDD benchmark dataset. They measure precision, recall and F1-score and compare with a linear regression model and an SVM model. They extract rules from the tree for explainability purposes, and conclude that decision trees provide a trustworthy and explainable model. They do not compare their work with deep learning models, and they do not provide any metrics to support their claims of increased trustworthiness and explainability.

B. MOTIVATION OF WORK

The purpose of this study is to explore whether it is possible to *increase the trustworthiness* of an IDS that monitors vehicle data, using XAI tools (SHAP in particular). The study does *not* aim to evaluate the performance of XAI powered AI models. Moreover, this study aims to provide results that evaluates the trustworthiness (and explainability) in a quantitative way (which is done by a user survey). This particular scope hasn't been addressed in the reviewed literature.

The general consensus seems to be that in order for AI to gain further widespread use, there must be an established trust in such technology that is convincing enough to allow manufacturers and companies to apply them in current real-world situations [34]. Such situations are those where the consequences of faulty predictions are critical, and where the functional opacity of the model is affected by its complexity [24]. AI-based IDSs in automotive cybersecurity fits that category of situations.

Because of the lack of interpretability in sophisticated AI-based, in-vehicle IDSs, it is difficult for humans to trust such systems, let alone know what actions to take when an IDS flags an attack. *By using explainability tools found in the area of XAI, this study aims to explore an area which is lacking research; what kind of explanations could be produced in accord with model predictions, to further increase the explainability, and by extension the trustworthiness, of AI-based IV-IDSs.*

C. SUMMARY OF CONTRIBUTIONS

Thus, we propose an XAI model for an AI-based IV-IDS, that increases its trustworthiness, hence, addressing the

forementioned issues. The main contributions of this paper are as follows:

- First, we perform a literature study (section I-A) of the state-of-the-art advancements in IDS research (with a particular focus on IV-IDSs). Our findings show that research on XAI in IV-IDSs is not well-explored. In particular, quantitative results of IV-IDSs' trustworthiness requires significant attention.
- Second, to enhance the explainability of an IV-IDS system, we develop and propose a visual explanation, named VisExp (section III). VisExp applies the state-of-the-art feature importance method "SHAP" [29] to show the behavior of the IV-IDS. In our experiment, a DNN-based IDS is used and trained on a tried-and-tested IDS dataset. We compare the outcome of VisExp with a rule-based explanation, which is a knowledge extraction technique (state-of-the-art).
- Third, using VisExp and the rule-based explanation, we conduct a user survey that quantitatively evaluates the trustworthiness of the XAI-powered IV-IDS system (section IV). The survey participants consist of experts with domain knowledge in areas such as AI, automotive, and cybersecurity.
- Lastly, we perform an extensive analysis of the survey results based on trustworthiness, explainability, and general trust (section V). The results show that the experts' trust in the proposed AI-based IV-IDS is increased when using explanations (e.g. VisExp) that explain the IDS system behavior.

II. PROPOSED APPROACH FOR AI-BASED IV-IDS

In this study, the detection of manipulated CAN frames (see Figure 1) by the IV-IDS is viewed as a binary classification task. The approach is to train a classification model with *training data* in order to predict the label (class) of "new" *test data*. Three datasets were looked at; "Survival" [13], "ROAD" [35], and "Hisingen" (proprietary), and they would all have been used virtually the same way. For the sake of this study, however, only one dataset was used; the "Survival" dataset, which is created by Han *et al.* [13] in their paper about using Survival analysis in IV-IDS design. The classes are "attack" (manipulated CAN frames) and "normal" (ordinary CAN frames). The dataset contains 1735840 instances of which 183314 are attacks. Because of its innate low degree of interpretability, a DNN was used (hyperparameters are shown in Table 1). The DNN is built using tensorflow/keras. It uses 3 fully connected hidden layers with rectified linear unit (ReLU) activation. A EarlyStopping callback is set up to stop the training when the loss no longer changes, and retrieve the model which performed best of all epochs. Kernel initialization method is `glorot_normal`. The loss function is `BinaryCrossentropy`.

Features were engineered (see Table 2) using the ID, data field, and timestamp features (see Figure 1) of the Survival dataset.

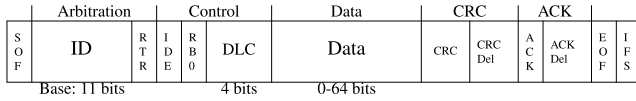


FIGURE 1. CAN frame | The Survival dataset has features of the ID, DLC and data field, along with the timestamp of when a CAN frame is transmitted.

TABLE 1. DNN hyperparameters | Parameters and their values as specified when building the DNN in keras.

Layer	# of units	Description
layer_1	11	keras.layers.Dense
layer_2	23	keras.layers.Dense
layer_3	7	keras.layers.Dense
Hyperparameter	Value	
optimizer	"adam"	Optimizer algorithm
batch_size	200	# of samples in a gradient descent
epochs	20	# of training passes over the dataset

TABLE 2. The engineered features.

Feature	Description
dt [12]	Transmission time (s) between CAN frames
dt_ID [12]	Transmission time (s) between CAN frames with the same ID
dt_data	Transmission time (s) between CAN frames with the same data field
dcs	Data change score (ratio) between CAN frames
dcs_ID	Data change score (ratio) between CAN frames with the same ID

Two of the features that were engineered are: *dcs* (data change score), which is the *change* in the data field since the last frame, and *dcs_ID*, which is the change in the data field since the last frame *with the same ID*. The change is calculated with the simple matching coefficient (SMC), which has the formula:

$$SMC = \frac{\text{number of matching bits in } d_1 \text{ and } d_2}{\text{number of bits in } d_1 \text{ (or in } d_2)} \quad (1)$$

where d_1 and d_2 are data fields in two different CAN frames. Since SMC is a similarity measure, to get the change, $1 - SMC$ is calculated. This produces a score between 0 (no change) and 1 (total change). The features in Table 2, along with the label (0 = "normal", 1 = "attack"), are used to train the DNN.

Training (70%) and testing (30%) the DNN results in 746 false positives (false alarms) and 536 false negatives. The true positive rate is 0.998 and the true negative rate is 0.9998. The DNN has an accuracy of 0.9973 (std = 0.0004) and an F1-score of 0.9868 (std = 0.0028) in a 10-fold cross validation using the training data.

III. PROPOSED APPROACH OF XAI METHOD: VisExp

This section explains the XAI method used to increase the trustworthiness of the IV-IDS described in Section II.

A. SHAP (SHapley ADDITIVE exPlanations)

At the core of VisExp is SHAP. It extracts the feature importance for each feature in predicted instances (CAN frame), which is utilized in VisExp. The following subsection is an overview of SHAP.

SHAP is a framework of explainability methods developed by Lundberg and Lee [36], that weaves together previously proposed methods, such as LIME [37], DeepLIFT [38], and Layer-Wise Relevance Propagation [39]. The authors classify these methods as being *additive feature attributions*, in that they have a similar structure to each other, and to Shapley values from coalitional game theory (which are also deemed as additive feature attributions). Additive feature attribution methods are linear functions of binary variables:

$$g(z') = \phi_0 + \sum_{i=1}^M \phi_i z'_i \quad (2)$$

where g is the explanation model, $z' \in \{0, 1\}^M$ is the coalition vector, M is the maximum coalition size, and $\phi_i \in \mathbb{R}$ is the feature attribution of feature i , i.e., the Shapley value [25]. The Shapley value for a feature in an instance is computed by first computing the difference of every *coalition*, with the feature included and not included. Next, taking the (weighted) average of all those differences amounts to the Shapley value for that feature in that instance. The method uses the fact that Shapley values guarantee solutions, and applies that in their framework, proposing *SHAP values* as solutions for their additive feature attribution driven method [29].

SHAP is an explainability method that uses feature importance, through simpler surrogate models [24] (KernelSHAP, TreeSHAP, DeepSHAP, etc.). KernelSHAP is one such model that is independent of the AI model, and is as such a model agnostic method, hence, it is used in this study (see Section IV). SHAP is also heavy on the visualization [24] aspect of explanations. Since it is based on Shapley values, which is calculated for each input instance being predicted, it is a local [24] explainability method. It does provide pseudo-global interpretations through aggregating SHAP values. It is a post-hoc [24] method, since it is applied on an AI model after it has been trained.

B. VisExp: A CUSTOM VISUALIZATION-BASED EXPLANATION

VisExp is a custom, pseudo-global, explanation that was created in order to increase the trustworthiness of, and for domain experts to gain insights about, the AI-based IV-IDS. It was created using well known python libraies, such as matplotlib and Seaborn. By generating SHAP values from the AI model (DNN) using a class balanced sample of training data (600 instances), a visualization-based explanation is

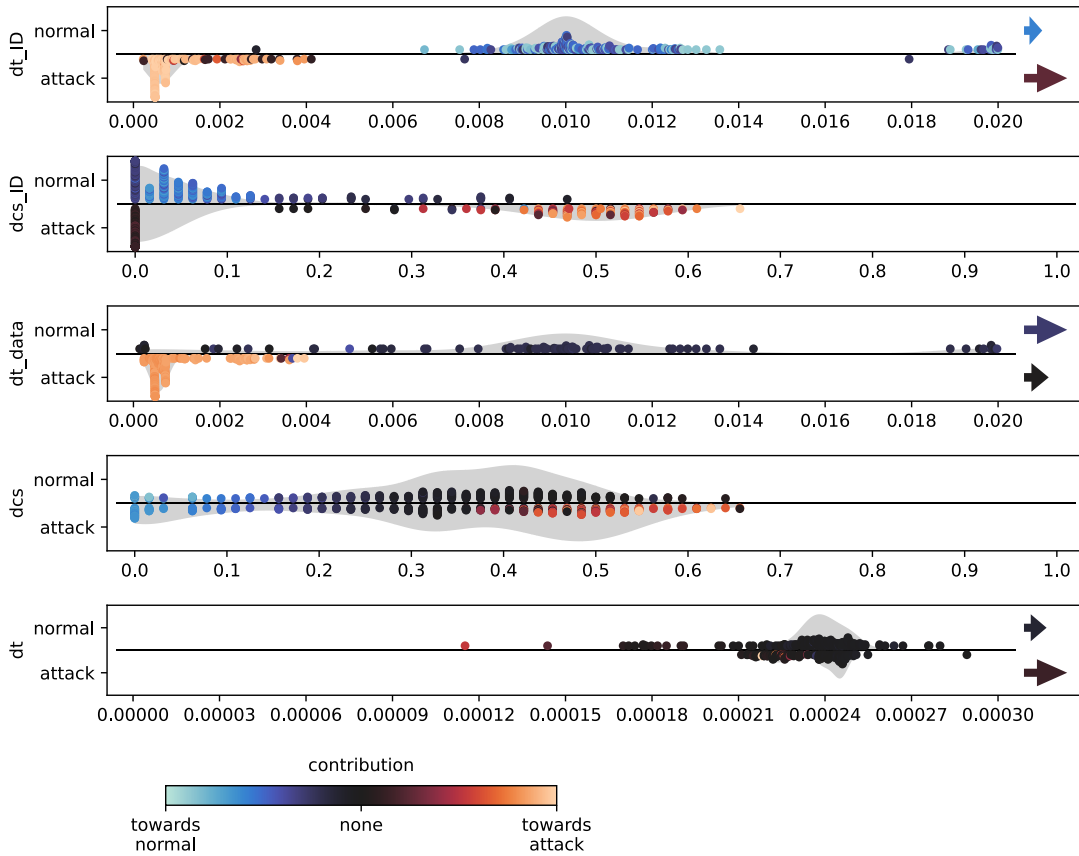


FIGURE 2. VisExp | A pseudo-global visualization-based explanation, using SHAP values. It shows the features in the dataset in swarm plot-like strips for normal and attack classifications. Each point is an instance from the train data. The x-axes are the feature values, and the color represents the SHAP values. The color of the arrows represent the mean of the SHAP values outside of the diagram, and their relative size represents how many data points there are.

developed (see Figure 2). The explanation is pseudo-global, in the sense that aggregated (combined in the plot) local explanations give insight about the global behavior of the AI model.

The structure of the explanation is as follows: For each feature, a dual swarm plot is generated, that shows the normal CAN traffic (in the training data) at the top, and the attack traffic at the bottom. A violin plot (Seaborn) is generated behind the swarm plots in order to further highlight the density of instances at each position. SHAP values are represented by colors (following a diverging color scheme), such that one end of the spectrum (e.g., light blue) represents low SHAP values and the other end (e.g., light orange) represents high SHAP values. In this way, it is possible to spot characteristics in contribution (blue vs orange) separate from actual predictions (normal vs attack) made by the model (it should be noted that this effect is not satisfyingly clear in the case of the dataset and features used in this study, alas, all characteristic values of features belonging to a class (e.g. attack) implicate consistent contribution (SHAP values) towards that class).

In the case where the range of a feature is very large, i.e., feature values are spread out across a large span, where there are important details in a shorter span, arrows of different

sizes and colors are plotted to represent how many instances are outside of the plot, and what contribution they have. A larger arrow represents more instances than a small arrow, and the color scheme is the same as for instances within the plotted range, but the color represents the mean contribution of all outliers.

C. CUSTOM RULE-BASED EXPLANATION

Drawing from the principles of knowledge extraction [24], a rule-based, textual, explanation was constructed to be compared with VisExp. It, as well as VisExp, uses predictions of the training data with the DNN model. The rule-based explanation was defined by setting thresholds that divides attacks and normal data in a rough fashion (by looking at VisExp in Figure 2). Only two features were considered, as its function is to compare with VisExp in the survey (Section IV). The rule-based explanation is shown in Figure 3.

IV. EVALUATING VisExp USING A QUANTITATIVE SURVEY

To evaluate the trustworthiness and explainability using VisExp (compared to the rule-based explanation), a human-grounded [23] experiment in the form of a survey was

It is often “normal” if
 dt_ID is between 0.005 and 0.02,
 dcs_ID is under 0.3

It is often “attack” if
 dt_ID is under 0.005 or over 0.02,
 dcs_ID is over 0.3

FIGURE 3. Rule-based explanation | Two features are explained. It explains the model behavior by defining thresholds where the AI model roughly transitions between normal and attack predictions.

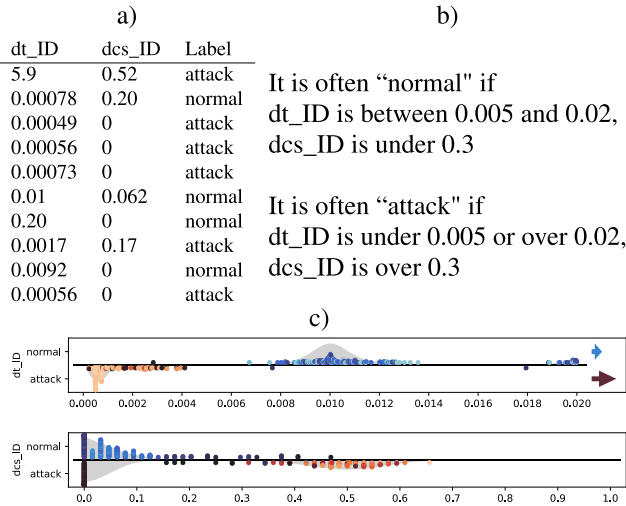


FIGURE 4. The explanations used in the survey | a) is a table consisting of inputs and outputs from the DNN model, b) is a rule-based explanation describing the relationships between classifications and values of dt_ID and dcs_ID, c) is VisExp, which shows previously predicted data points by the DNN, along with SHAP values represented by color.

created. The survey aims to capture the experts’ trust in, and understanding of, an advanced AI-based IV-IDS. The specific AI-model used (DNN) is not told to the participants. It was trained on all the attack sub-datasets from the Survival dataset. The reason it was trained on all attacks is to gain a variety of attacks in the model, and by extension in the explanations provided to the participants. There are three explanations used in the survey (see Figure 4; one is only model (DNN) input/output, one is the rule-based explanation, and one is VisExp (see Section III-B). Because of the fact that KernelSHAP can be used on any model (i.e., being model agnostic [29]), it was chosen to generate the SHAP values used in VisExp and the rule-based explanation. The survey was mailed to domain experts in AI, cybersecurity and automotive. The domain experts are mainly researchers at Mid Sweden University and RISE, all participated willingly.

For the input and output of the DNN model (see Table 4a), 10 random samples are extracted from the test set of the Survival dataset, and run through the model, generating outputs seen in the table. In preparation of the survey, some features were omitted, and only 2 features were included in the experiment; dt_ID and dcs_ID. Although the preprocessed dataset contains 5 features, two of the most important ones are dt_ID

TABLE 3. Survey questions | This is how the questions were asked in the survey, with the “Visual explanation” being VisExp, the “Textual explanation” being the rule-based explanation, and the “Table” being previous model input/output.

#	Question
1	“I trust the decisions generated from Artificial Intelligence for vehicle safety.”
2	The advanced Artificial Intelligence security program indicates that your vehicle has been attacked. How much do you agree with the following statement: “I trust the security program’s judgment.”
3	The advanced Artificial Intelligence security program indicated that your vehicle has been attacked, because feature dt_ID’s value is 1.5 and feature dcs_IDs value is 0.61...
4	...Please, look at the Table below, which describes previous judgments made by the security program. How much do you agree with the following statement: “The Table assured me, leading me to trust the security program’s judgment.”
5	...Please, look at the Textual explanation below, which describes how the security program makes judgments. How much do you agree with the following statement: “The Textual explanation assured me, leading me to trust the security program’s judgment.”
6	...Please, look at the Visual explanation below, which describes how the security program makes judgments. How much do you agree with the following statement: “The Visual explanation assured me, leading me to trust the security program’s judgment.”
7	Which one enhances your understanding of Artificial Intelligence decision-making the most? Choose an option below.
8	How much do you agree/disagree with this statement: “The Visual explanation is more trustworthy and understandable than the Textual explanation.”
9	How much do you agree/disagree with this statement: “The Visual explanation is more revealing and descriptive than the Textual explanation.”

and dcs_ID, as they alone, using the rules in Figure 4b, can unofficially predict the correct class with an accuracy of about 90%. Further, when the input is presented to the participants, dt_ID and dcs_ID both have a maximum of 2 significant digits. The Label column, formerly consisting of 0s for “normal” and 1s for “attack”, has been changed to strings of “normal” and “attack” for readability (refer to Figure 4a to view the format).

The general format of the questions are statements (see Table 3), to which the participants answer how much they agree/disagree, on a Likert scale [40], from 1 to 5. When the answers are processed, i.e., means and standard deviations are calculated (as presented in the results in Section V), they are numbered from 0 to 4. The participants are asked to answer questions relating to if they would trust the IV-IDS, if provided with different explanations (of those in Figure 4). One question is a *binary forced choice* [23], where the participants are faced with VisExp and the rule-based explanations, and must answer which one they feel enhances their understanding of the AI model. The structure of the survey is depicted in Figure 5.

V. SURVEY RESULTS AND ANALYSIS

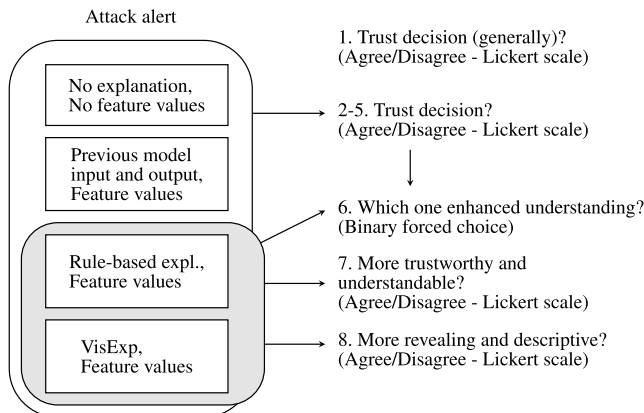
The survey received 30 answers. The participants came from a variety of expertise areas within computer engineering, most notably, AI, cybersecurity, and wireless communication (see Table 6). They are divided into Primary and Secondary Experts where Primary Experts are the AI-, Automotive- and Cybersecurity Experts; the Secondary Experts are the

TABLE 4. Survey highlights | Q = Questions.

Evaluation	Q	Results
Trustworthiness	3-5, 7	Primary Experts (and all participants) trust the IV-IDS with VisExp the most (Table 5). The majority of all participants believes that VisExp is more trustworthy and understandable than the rule-based explanation (Figure 7).
Explainability	6, 8	Primary Experts (and all participants) think that VisExp enhanced their understanding of AI decision-making the most (Figure 6). Most participants think VisExp is more revealing and descriptive than the rule-based explanation (Figure 8).
General trust	1, 2	The answers to Question 1 indicate that there is some division regarding decisions made by an AI system for vehicle safety (51.7% would trust it, while 31% would not). Question 2's answers show a stronger indication towards trust (65.5% would trust it, 10.3% would not).

TABLE 5. Survey answers | Mean and standard deviation of answers to all Likert style questions. Interesting results are bolded. Q = "Question"; RB = rule-based explanation; tu = "trustworthy and understandable"; rd = "revealing and descriptive":

Expertise area	Mean(Standard deviation)						Q7 (tu) "VisExp>RB"	Q8 (rd) "VisExp>RB"
	Q1 "General"	Q2 "Alerted"	Q3 "In-/output"	Q4 "Rule-based"	Q5 "VisExp"			
All participants	2.233(0.920)	2.600(0.757)	1.967(1.048)	2.367(0.983)	2.500(0.992)		2.567(1.116)	2.967(0.983)
Primary Experts	2.071(0.884)	2.357(0.718)	2.000(0.926)	2.357(0.811)	2.786(0.860)		2.643(0.972)	3.214(0.860)
Secondary Experts	2.375(0.927)	2.813(0.726)	1.938(1.144)	2.375(1.111)	2.250(1.031)		2.500(1.225)	2.750(1.031)
AI Novice	2.231(0.890)	2.692(0.722)	1.923(1.071)	2.231(1.120)	2.231(0.973)		2.000(1.301)	2.615(1.146)
AI Expert	2.250(0.829)	2.500(0.707)	2.250(0.661)	2.500(0.866)	3.000(0.707)		3.125(0.599)	3.375(0.696)
Cybersecurity	1.857(0.833)	2.143(0.639)	1.714(1.030)	2.286(0.700)	2.571(0.904)		2.143(0.990)	3.143(0.990)
Wireless/IoT/Comm.	2.333(1.106)	2.667(0.745)	1.833(1.213)	2.167(0.898)	2.500(1.118)		3.167(0.373)	3.167(0.687)
Data Analysis Expert	1.750(0.829)	1.750(0.829)	1.500(0.866)	2.750(0.433)	2.750(1.090)		3.000(1.225)	3.250(0.829)
Automotive Expert	2.000(0.816)	2.000(0.816)	2.333(0.471)	3.000(0.000)	3.667(0.471)		3.333(0.471)	4.000(0.000)

**FIGURE 5.** Survey structure | Question 1 is a general question about trust in AI for automotive safety. Question 2 asks about trust when the IV-IDS alerts an attack. Questions 3-5 include all explanations (Figure 4), and leads to Question 6, and then Questions 7-8, which all include VisExp and the rule-based explanation.

Wireless/IoT/Communication- and Data Analysis Experts and AI Novices. Highlights of the results can be seen in Table 4, and an overview of answers to the different explanations can be seen in Figure 6. Answers to all questions are in A, Table 7.

A. TRUSTWORTHINESS: QUESTIONS 3-5, 7

Looking at **Questions 3-5** (Table 3), the results in Table 5; Q3-5 show that Primary Experts' trust in the IV-IDS is significantly stronger when it is explained with VisExp (mean = 2.786 (Agree), std = 0.860), than when it is explained through

TABLE 6. Expertise areas |. Note that participants could choose more than one area, hence, the percentages do not add up to 100%, i.e., they are percentages of the total population (30). The Primary Experts are the AI-, Automotive- and Cybersecurity Experts; the Secondary Experts are the Wireless/IoT/Comm.- and Data Analysis Experts and AI Novices.

Expertise area	Count	% of population (30)
Primary Experts	14	46.7%
Secondary Experts	21	70%
AI Novice	13	43.3%
AI Expert	8	26.7%
Cybersecurity	7	23.3%
Wireless/IoT/Comm.	6	20%
Data Analysis Expert	4	13.3%
Automotive Expert	3	10%

previous model input and output (mean = 2.000 (Undecided), std = 0.926), with a mean difference of 0.786 (~ 1 step up on the Likert scale [40]; two-tailed paired $t(26) = 6.904$, $p < 0.001$). Compare this with the mean difference between the rule-based explanation (mean = 2.357 (Undecided), std = 0.811) and the previous model input/output (mean = 2.000 (Undecided), std = 0.926), which is 0.357 (two-tailed paired $t(26) = 2.687$, $p = 0.019$), i.e., significant but less so.

Moreover (as seen in Table 5; Q5), Primary Experts trust VisExp (mean = 2.786 (Agree), std = 0.860) more than Secondary Experts (mean = 2.250 (Undecided), std = 1.031), with a mean difference of 0.536 (barely 1 step up on the Likert scale [40]; two-tailed unpaired $t(28) = 1.481$, $p = 0.15$).

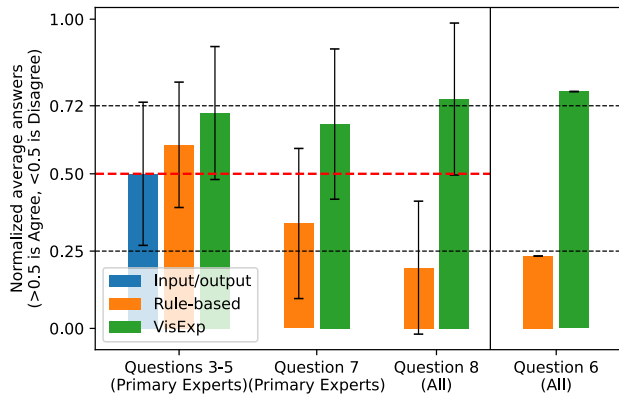


FIGURE 6. Normalized averages of the answers to all questions with explanations | Questions 3-5 and 7 are answers from Primary Experts. Questions 8 and 6 are answers from all participants. Question 6 is a binary forced choice, and does not abide by the red agreement threshold.

Between VisExp (Table 5; Q5) and the rule-based explanation (Table 5; Q4), Primary Experts' trust in the IV-IDS is significantly stronger with VisExp, with a mean difference of 0.429 (two-tailed paired $t(26) = 3.122, p = 0.0081$). For Secondary Experts, the mean difference between VisExp and the rule-based explanation is less, and the rule-based explanation has a slightly higher mean.

Looking at all participants (see Table 5), the results show that their trust in the IV-IDS is significantly stronger when it is explained with VisExp (mean = 2.500 (Agree), std = 0.992), than when it is explained with the rule-based explanation (mean = 2.367 (Undecided), std = 0.983), with a mean difference of 0.113 (two-tailed paired $t(58) = 2.112, p = 0.043$).

Looking at **Question 7** (Table 3), the majority of all participants think that VisExp is more trustworthy and understandable than the rule-based explanation (72.4% in agreement vs. 27.5% in disagreement, see Figure 7). AI-, Wireless/IoT/Comm.-, Data Analysis- and Automotive Experts all think that VisExp is more trustworthy and understandable than the rule-based explanation (see Table 5; Q7).

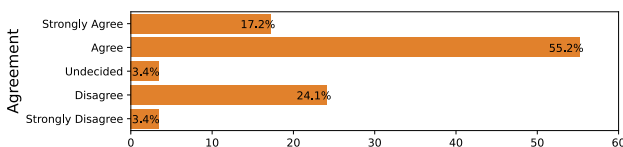


FIGURE 7. Question 7 answers.

B. EXPLAINABILITY: QUESTIONS 6, 8

Looking at **Question 6** (Table 3), 77% thought that VisExp enhanced their understanding the most, while the remaining 23% thought the rule-based explanation enhanced their understanding the most (see Figure 6; Question 6).

Looking at **Question 8** (Table 3), there is an overwhelming majority (79.3%, of which 43.5% (34.5% of total) is Strong Agreement) of all participants that thinks that VisExp is more

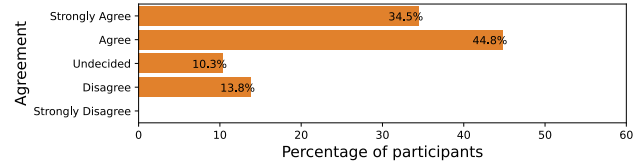


FIGURE 8. Question 8 answers.

revealing and descriptive than the rule-based explanation (see Figure 8). The only ones who seem to disagree are the AI Novices (see Table 5; Q8).

C. GENERAL TRUST: QUESTIONS 1, 2

On **Question 1** (Table 3), the majority were positive (51.7%), saying that they would trust the program's decisions. However, there was a relatively large portion who were in disagreement (31%) (see Figure 9a). **Question 2** (Table 3) asks if participants would trust the AI if, in the vehicle, it alerts the driver that their vehicle has been attacked. This question is similar to the first, but yields different results, and the answers show that a larger majority is positive (65.5%), and that there are fewer who are in disagreement (10.3%) (see Figure 9b).

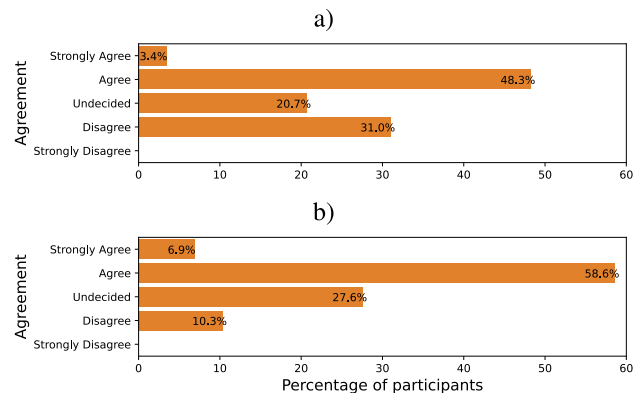


FIGURE 9. Question 1 and 2 answers | a) Question 1 and b) Question 2. Even though the questions are similar, the answers are very different.

VI. DISCUSSION

The way of “seeing” the behavior of the AI, as is done with VisExp, shows information about where predictions are likely to end up, by referring to how the model predicts data it has already trained with. The information gained by studying/looking at the explanation, while difficult to quantify, is likely higher than without an explanation (which is supported by the results in Section V).

The answers to Question 7—that VisExp is more trustworthy and understandable than the rule-based explanation—correlates with the answers to Question 6—that VisExp enhances the participants' understanding of AI decision-making the most. This could point towards a positive correlation between the understandability and trustworthiness of an explanation. It was unofficially hypothesized that the rule-based explanation would get more votes in terms of

understandability and trustworthiness, because of its simplicity. In the beginning of the survey, as answers were still coming in, this looked like the trend.

A. OPEN ISSUES

When developing VisExp, the training data was used to represent the behavior of the AI-based IV-IDS. Instead of using all the predicted/true labels of the training data, only using TPs and TNs could be studied and explored. This way, there will be no misclassifications in the visualization, showing more of what the model “does right”. However, keeping misclassifications does have the benefit of showing the models *truthful* behavior, which is crucial when learning why a model makes mistakes, hence, studies could instead be made in an effort to implement a visual representation of misclassified instances. Another thing that can be done is to highlight a prediction made by the IV-IDS in the VisExp plot interactively, indicating the class visually in VisExp.

It could be argued that the way the survey was formatted on Google Forms might skew the results—if the answers should have been independent of each other. Since the participants answered questions about the rule-based explanations before VisExp, they might have combined insights from all explanations when answering. In a future study, the other explanations might be hidden from participants, as they answer questions regarding the current one. Further, forward simulation [23] experiments, as proposed by [41], where participants are tasked to guess the output of an AI model by looking at the input, can be utilized to get a score of how effectively an explanation enhances people’s understanding of the model.

The number of participants (30) in the survey might change the results. There are significant findings in the current experiment, but more participants of the same quality (domain experts) might give more accurate results. The results show a difference between the primary and the secondary experts’ trust towards the IV-IDS, and further experiments with these two groups might highlight the differences further.

A shortcoming of this study work is the lack of variety in the XAI tools studied—except for the different SHAP explainers. There are many XAI methods out there that could be utilized for classification tasks such as in IV-IDSs. One such tool is layer-wise relevance propagation. While it is mostly applied to AI tasks involving images—to see saliency maps of important pixels—there are implementations that use it for tabular data. Another interesting thing to study would be self-attention methods, and further apply natural language processing (NLP) concepts to tabular classification tasks, such as previously mentioned tokenization or vector embeddings.

The methods that generate SHAP values in the SHAP framework are approximations of true Shapley values, and as such there is always an error. The official SHAP implementation does not include the error bounds of the Shapley

TABLE 7. Raw survey answers | Questions 3-5, 7, 8: 0 = “Strongly Disagree”, 1 = “Disagree”, 2 = “Undecided”, 3 = “Agree”, 4 = “Strongly Agree”; Question 6: 0 = “rule-based explanation”, 1 = “VisExp”.

Expertise area	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
AI Expert, Data Analysis Expert, Automotive Expert	3	1	2	3	4	1	4	4
Data Analysis Expert	1	1	2	3	3	1	4	4
AI Expert, Automotive Expert, Cybersecurity	2	2	2	3	3	1	3	4
AI Novice	3	3	1	3	1	0	0	1
AI Novice, Cybersecurity	1	1	0	1	1	0	1	1
Wireless	4	4	0	3	4	1	3	4
AI Expert, Wireless	1	2	1	2	2	1	3	3
AI Novice, Cybersecurity	1	2	2	2	2	0	1	3
Wireless	3	3	3	1	2	1	3	3
AI Novice	3	3	1	2	2	1	3	3
Cybersecurity	3	3	3	3	3	1	3	3
Data Analysis Expert	1	2	0	3	1	0	1	2
AI Expert	3	3	2	2	2	1	2	2
AI Expert	2	3	3	4	3	1	3	4
Automotive Expert, Cybersecurity	1	3	3	3	4	1	3	4
AI Novice, Cybersecurity	3	2	1	2	3	1	1	4
AI Novice	1	3	2	3	1	0	1	4
AI Expert	3	3	3	3	3	1	3	3
AI Expert	3	3	2	1	4	1	4	4
AI Novice	3	3	4	4	1	0	1	1
AI Novice	2	3	3	0	3	1	3	3
AI Novice, Wireless	1	2	3	3	4	1	4	4
AI Expert	1	3	3	2	3	1	3	3
Wireless	2	2	1	1	1	1	3	3
Data Analysis Expert, AI Novice	2	3	2	2	3	1	3	3
AI Novice	3	4	3	4	2	0	1	1
AI Novice	3	3	1	1	3	1	3	3
AI Novice	3	3	2	2	3	1	4	3
Wireless	3	3	3	3	2	1	3	2
Cybersecurity	2	2	1	2	2	1	3	3

approximations [42]. This can skew the generated SHAP values, and give undesirable feature contribution values. In future projects, improvements to the SHAP framework can be made, that takes error bounds into account.

B. ETHICAL CONSIDERATIONS

The survey was done anonymously as per General Data Protection Regulation (GDPR). No personal information was saved, only the answers were saved. Furthermore, since a survey involves asking questions repeatedly, the number of questions was adapted so that the survey would not take too long. It was assumed that it would take at most 5-6 minutes.

VII. CONCLUSION

In this work, we addressed the challenge of deploying an AI-based IV-IDS from the perspective of trustworthiness and explainability. Through a creative process, and with the help of various python libraries and SHAP, a visualization-based explanation, VisExp, was created. VisExp shows the behavior of the AI-based IV-IDS, through swarm-like plots for each feature, highlighting the “attack” and “normal” classifications, along with feature and SHAP values, of the training data. From the survey analysis, questions related to the trustworthiness and explainability of the AI-based IV-IDS was answered by 30 experts within automotive, cybersecurity, AI, and other computer engineering fields. The survey comparatively questions VisExp along with a rule-based explanation, and the results show that VisExp significantly (i) enhanced participants’ understanding of, (ii) increased the trustworthiness of, and (iii) heightened the explainability of the IV-IDS. It did this to a higher degree than the rule-based explanation. This indicates that using explainability tools, such as VisExp,

can increase experts' trust towards sophisticated AI-based IV-IDSs, by explaining how it makes its decisions.

APPENDIX A SURVEY DETAILS

See Table 7.

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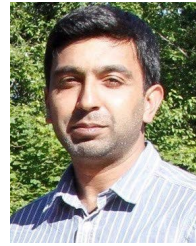
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